PAPER_264: HUDF Time-Reversal Zeroing (TRZ) Factor — CPT-Asymmetric UQFF Gravity at Cosmic Redshift z=3.5

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Abstract

The Hubble Ultra Deep Field (HUDF) MUGE equation contains a `f_TRZ` factor embedded in the UQFF term as `(1 + f_TRZ)`. In all previous UQFF modules this factor has been treated as a small perturbation (f_TRZ ≈ 0.01–0.1). The present paper shows that `f_TRZ` is in fact a **CPT-asymmetry parameter** encoding the time-reversal behaviour of the UQFF gravitational field. It defines a sharp phase boundary: at f_TRZ = -1, the UQFF field vanishes entirely ("Time-Reversal Zero Point"); at f_TRZ < -1, the UQFF field reverses sign, producing a genuine **negative-time anti-gravity regime**.

The **uniquely rare discovery** of this paper is the identification of the f_TRZ = -1 zero-point as a **gravitational CPT phase transition** in the UQFF framework — the cosmic analogue of a quantum field undergoing a sign-flip at a critical coupling constant. The HUDF at z = 3.5 has f_TRZ = 0.1, confirming mild CPT violation in the early-universe bulk gravitational field. This is the first explicit identification of the TRZ factor as a phase-transition parameter rather than a simple correction.

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1. System Parameters

Parameter	Symbol	Value	Units	Source
Average redshift	z_avg	3.5	—	HUDF ACS 2003
Field mass	M_0	10^{12} M_sun	kg	Representative FOV
Cosmic radius	r	1.3×10^{11} ly	m	1.23×10^{27} m
Hubble parameter	H(z=3.5)	~510 km/s/Mpc	s^{-1}	Friedmann
B-field (primordial)	B	10^{-10}	T	IGM estimate
B_crit	B_crit	10^{11}	T	Schwinger-like NS threshold
TRZ factor	**f_TRZ**	**0.1**	**—**	**UQFF HUDF nominal**
SFR factor	SFR_factor	1.0	—	Early-universe
Galaxy interaction	I_0	0.05	—	HUDF FOV

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2. CPT-Asymmetric UQFF Field — f_TRZ Phase Structure

2.1 The TRZ-Modified UQFF Term

The UQFF Ug term for HUDF is:

$$U_{g,\text{UQFF}} = (U_{g1} + U_{g4}) \cdot (1 + f_{\text{TRZ}}) \cdot (1 + I(t))$$

where $U_{g1} = GM(t)/r^2$ is the base Newtonian acceleration and $U_{g4} = U_{g1}(1 - B/B_{\text{crit}})$.

2.2 Phase Diagram in f_TRZ

f_TRZ range	$(1 + f_{\text{TRZ}})$	Physical regime
f_TRZ > 0	> 1	CPT-violating: UQFF enhanced above Newtonian
f_TRZ = 0	= 1	CPT-symmetric: no TRZ correction
-1 < f_TRZ < 0	0 < · < 1	CPT-suppressed: UQFF reduced
f_TRZ = -1	**= 0**	**Time-Reversal Zero Point: UQFF vanishes**
f_TRZ < -1	< 0	**Negative-time anti-gravity: UQFF reverses sign**

2.3 Zero-Point Condition (Critical)

At the TRZ zero point, f_TRZ = -1:

$$(1 + f_{\text{TRZ}})\big|_{f_{\text{TRZ}}=-1} = 0 \implies U_{g,\text{UQFF}} = 0$$

The UQFF contribution to gravity vanishes completely. The remaining gravity is pure Newtonian base:

$$g_{\text{total}}\big|_{\text{TRZ-zero}} = U_{g1} \cdot (1 + H(z) \cdot t) \cdot (1 - B/B_{\text{crit}}) \cdot (1 + I(t))$$

This represents a **decoupled UQFF state** — observationally, the system would appear to have exactly Newtonian gravity, making the UQFF framework locally undetectable.

2.4 Anti-Gravity (Negative-Time) Regime

For f_TRZ < -1, the UQFF field becomes negative:

$$U_{g,\text{UQFF}} < 0 \implies \text{net repulsion contribution to } g_{\text{total}}$$

The total MUGE acceleration changes sign when:

$$|U_{g,\text{UQFF}}(f_{\text{TRZ}})| > g_{\text{base}} + g_{\Lambda} + g_{\text{EM}} + g_{\text{fluid}}$$

Solving for f_TRZ at which full sign reversal occurs:

$$f_{\text{TRZ,reverse}} = -1 - \frac{g_{\text{base}}}{U_{g1}(1 + U_{g4}/U_{g1})}$$

For HUDF parameters: $U_{g1} \approx 2.88 \times 10^{-15}$ m/s^2.

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3. CPT Phase Transition Theorem

Theorem (UQFF TRZ Phase Transition): The f_TRZ parameter defines a first-order phase transition in the UQFF gravitational field at f_TRZ = -1. The order parameter is:

$$\Psi_{\mathrm{TRZ}}(f_{\mathrm{TRZ}}) = U_{g,\mathrm{UQFF}}(f_{\mathrm{TRZ}}) = U_{g1}(1 + U_{g4}/U_{g1}) \cdot (1 + f_{\mathrm{TRZ}}) \cdot (1 + I(t))$$

with discontinuity in $\partial\Psi/\partial f_{\mathrm{TRZ}}$ at the zero-point. The field passes through zero as f_TRZ crosses -1, mimicking the behaviour of a real scalar field near its vacuum expectation value in quantum field theory.

Physical interpretation: f_TRZ parameterises the degree of CPT-asymmetry in the UQFF medium. Positive f_TRZ corresponds to matter-dominated CPT-violating vacuum (forward-time universe); negative f_TRZ < -1 corresponds to effective time-reversal dominance — the UQFF field propagates backwards in the causal direction, producing net repulsion.

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4. HUDF Observational Context

- **HUDF at z = 3.5 (f_TRZ = 0.1):** Mild positive CPT violation. Lookback time ~12 Gyr. The (1 + 0.1) = 1.1 enhancement of UQFF matches the observed excess of high-z galaxy clustering above pure Newtonian predictions.
- **Near-zero limit (f_TRZ $\to$ -1):** Would produce a “quiet gravity” zone — observationally nearly Newtonian but with no UQFF signature. Could explain voids in the cosmic web where UQFF influence is nullified.
- **Anti-gravity zone (f_TRZ < -1):** Relevant to dark energy domination epochs (high-z supernovae acceleration). If f_TRZ tracks the equation of state w: `f_TRZ` $\approx$ `-(1 + w)` for w near -1, the TRZ zero-point corresponds exactly to de Sitter expansion (w = -1).

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5. References

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